

# TOPOLOGICAL GAMES AND BAIRE SPACES

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## 0. BAIRE CATEGORY

**Definition 0.1.** Let  $X$  be a topological space, and let  $E \subseteq X$ . The closure of  $E$  will be denoted by  $\overline{E}$ , and its interior by  $E^\circ$ .

- (1) The set  $E$  is said to be *nowhere dense* in  $X$  if  $(\overline{E})^\circ = \emptyset$ . Equivalently,  $E$  is nowhere dense if every non-empty open set  $U \subseteq X$  contains a non-empty open set  $V$  such that  $V \cap E = \emptyset$ .
- (2) The set  $E$  is said to be of the *first category* in  $X$  (or *meager*) if  $E$  is the union of a countable family of nowhere dense sets. A set which is not of the first category is said to be of the *second category* (or *non-meager*).
- (3) A set  $C \subseteq X$  is said to be *comeager* (or *residual*) if its complement  $X \setminus C$  is of the first category.
- (4) The space  $X$  is called a *Baire space* if the intersection of every countable family of dense open subsets of  $X$  is dense in  $X$ .



**Remark 0.2.** The notions in Definition 0.1 classify subsets according to topological “size,” providing a qualitative analogue of measure theory:

- (1) A nowhere dense set is essentially “full of holes.” It cannot contain any open neighborhood, nor can its points cluster thickly enough to fill any open region. For example, any finite subset of  $\mathbb{R}$  is nowhere dense, as is the Cantor ternary set.
- (2) A set of the first category is topologically negligible. A countable union of such negligible sets remains of the first category, just as a countable union of sets of Lebesgue measure zero has measure zero. The rational numbers  $\mathbb{Q} = \bigcup_{q \in \mathbb{Q}} \{q\}$  form a set of the first category in  $\mathbb{R}$ .
- (3) A comeager set represents an overwhelming majority of the space; its elements constitute the “generic” points of  $X$ . In  $\mathbb{R}$ , the set of irrational numbers is comeager.
- (4) A space is Baire if it cannot be thinned out into a negligible set of itself. In such a space, dense open sets are so large that countably many of them can be intersected without tearing open voids or depleting the space, ensuring that limit processes and approximations do not collapse into the empty set.

**Theorem 0.3 (S. Banach, 1935).** *A topological space  $X$  is a Baire space if and only if no non-empty open subset of  $X$  is of the first category.*

*Proof.* Suppose  $X$  is a Baire space, and let  $G \subseteq X$  be open and non-empty. If  $G = \bigcup_{n=1}^{\infty} E_n$  with each  $E_n$  nowhere dense, then  $V_n = X \setminus \overline{E_n}$  is open and dense in  $X$ . Hence  $\bigcap_{n=1}^{\infty} V_n$  is dense in  $X$ , so it must intersect  $G$ . However,

$$G \cap \bigcap_{n=1}^{\infty} V_n = G \setminus \bigcup_{n=1}^{\infty} \overline{E_n} = \emptyset,$$

a contradiction. Thus  $G$  is of the second category.

Conversely, suppose no non-empty open subset of  $X$  is of the first category. Let  $\{V_n\}$  be a sequence of dense open subsets of  $X$ , and let  $W \subseteq X$  be open and non-empty. Write  $E_n = X \setminus V_n$ . Since  $V_n$  is dense,  $(\overline{E_n})^\circ = (X \setminus V_n)^\circ = \emptyset$ , so each  $E_n$  is nowhere dense. If  $W \cap \bigcap_{n=1}^\infty V_n = \emptyset$ , then  $W \subseteq \bigcup_{n=1}^\infty E_n$ . This makes  $W$  a non-empty open set of the first category, contrary to our hypothesis. Hence  $W \cap \bigcap_{n=1}^\infty V_n \neq \emptyset$ , and the intersection is dense. 

## 1. INFINITE GAMES

**Definition 1.1.** Let  $A$  be a non-empty set. We denote by  $A^{<\omega}$  the collection of all finite sequences  $s = (a_0, \dots, a_{n-1})$  of elements of  $A$ . The length  $n$  of  $s$  is denoted by  $|s|$ ; the empty sequence  $\emptyset$  has length 0. If  $t \in A^{<\omega}$  and  $|t| \leq |s|$  with  $t_i = s_i$  for  $0 \leq i < |t|$ , we write  $t \subseteq s$ .

A non-empty subset  $T \subseteq A^{<\omega}$  is called a *tree* if  $s \in T$  and  $t \subseteq s$  implies  $t \in T$ . For each  $s \in T$ , we write

$$\text{Succ}(s) = \{a \in A : (s_0, \dots, s_{|s|-1}, a) \in T\}.$$

We assume throughout that  $\text{Succ}(s) \neq \emptyset$  for every  $s \in T$ . An infinite sequence  $x = (a_n)_{n=0}^\infty \in A^\omega$  is a *play* in  $T$  if  $(a_0, \dots, a_{n-1}) \in T$  for all  $n \geq 1$ . The set of all such plays is denoted by  $[T]$ .

**Definition 1.2.** An *infinite game* is a pair  $(T, W)$ , where  $T \subseteq A^{<\omega}$  is a tree and  $W \subseteq [T]$ . Two players, I and II, alternately choose elements  $a_n \in A$  such that  $(a_0, \dots, a_n) \in T$ . Player I chooses  $a_{2n}$ ; Player II chooses  $a_{2n+1}$  ( $n = 0, 1, 2, \dots$ ). Player II *wins* the resulting play  $x = (a_n) \in [T]$  if  $x \in W$ ; otherwise, Player I wins.

A *strategy* for Player I is a function  $\sigma$  defined on sequences of even length in  $T$  such that  $\sigma(s) \in \text{Succ}(s)$ . A play  $x$  is *consistent* with  $\sigma$  if  $a_{2n} = \sigma(a_0, \dots, a_{2n-1})$  for all  $n \geq 0$ . We say  $\sigma$  is a *winning strategy* for Player I if every play consistent with  $\sigma$  lies in  $[T] \setminus W$ .

A strategy  $\tau$  for Player II and the notion of a winning strategy for Player II are defined analogously.

## 2. THE BANACH–MAZUR GAME

**Definition 2.1.** Let  $X$  be a topological space, and let  $\tau_X^*$  be the collection of all non-empty open subsets of  $X$ . The *Banach–Mazur game* on  $X$ , denoted by  $\text{BM}(X)$ , is the game  $(T, W)$  where:

- (1)  $A = \tau_X^*$ .
- (2)  $T$  consists of all finite decreasing sequences  $U_0 \supset V_0 \supset U_1 \supset V_1 \supset \dots$  of sets in  $\tau_X^*$ .
- (3)  $W = \{(U_0, V_0, U_1, \dots) \in [T] : \bigcap_{n=0}^\infty U_n \neq \emptyset\}$ .

**Theorem 2.2.** *Let  $X$  be a topological space.*

- (1) *Player I has a winning strategy in  $\text{BM}(X)$  if and only if  $X$  contains a non-empty open set of the first category.*
- (2)  *$X$  is a Baire space if and only if Player I has no winning strategy in  $\text{BM}(X)$ .*

*Proof.* By Theorem 0.3, part (2) is an immediate consequence of part (1).

Suppose first that  $G \subseteq X$  is open, non-empty, and of the first category. Then  $G = \bigcup_{n=0}^\infty E_n$ , where each  $E_n$  is nowhere dense. Player I plays  $U_0 = G$ . If Player II responds with  $V_0 \subseteq U_0$ , the set  $V_0 \setminus \overline{E_0}$  is open and non-empty. Player I may therefore choose a non-empty open set  $U_1 \subseteq V_0 \setminus \overline{E_0}$ . In general, given  $V_{n-1}$ , Player I chooses an open set

$$U_n \subseteq V_{n-1} \setminus \overline{E_{n-1}}, \quad U_n \neq \emptyset.$$

This defines a strategy  $\sigma$  for Player I. For any play consistent with  $\sigma$ , we have  $U_n \cap E_{n-1} = \emptyset$  for all  $n \geq 1$ , so that

$$\bigcap_{n=0}^{\infty} U_n \subseteq G \setminus \bigcup_{n=0}^{\infty} E_n = \emptyset.$$

Thus  $\sigma$  is a winning strategy for Player I.

Conversely, suppose Player I has a winning strategy  $\sigma$ . Put  $U_0 = \sigma(\emptyset)$ . Let  $\Omega_0$  be a maximal collection of pairwise disjoint non-empty open subsets of  $U_0$ . For each  $V \in \Omega_0$ , Player I responds with  $U(V) = \sigma(U_0, V) \subseteq V$ . Let  $\Omega_1(V)$  be a maximal collection of pairwise disjoint non-empty open subsets of  $U(V)$ , and let  $\Omega_1 = \bigcup_{V \in \Omega_0} \Omega_1(V)$ . Continuing in this way, we obtain, for each  $n \geq 0$ , a collection  $\Omega_n$  of pairwise disjoint open sets such that  $D_n = \bigcup_{V \in \Omega_n} V$  is dense in  $U_0$ .

If  $U_0$  were of the second category, the intersection  $\bigcap_{n=0}^{\infty} D_n$  would be dense in  $U_0$ , hence non-empty. Choose  $p \in \bigcap_{n=0}^{\infty} D_n$ . There exists a unique sequence  $V_n \in \Omega_n$  such that  $p \in V_n$  for all  $n$ . This sequence defines a play consistent with  $\sigma$  in which  $p \in \bigcap_{n=0}^{\infty} U_n$ . This contradicts the assumption that  $\sigma$  is a winning strategy. Hence  $U_0$  is of the first category, and the proof is complete. 

### 3. THE STRONG CHOQUET GAME

**Definition 3.1.** Let  $X$  be a topological space. The *strong Choquet game* on  $X$ , denoted by  $\text{Ch}(X)$ , is played as follows. Player I chooses a pair  $(x_0, U_0)$  with  $x_0 \in U_0$  and  $U_0 \in \tau_X^*$ . Player II responds with  $V_0 \in \tau_X^*$  such that  $x_0 \in V_0 \subseteq U_0$ . At step  $n \geq 1$ , Player I chooses  $(x_n, U_n)$  such that  $x_n \in U_n \subseteq V_{n-1}$ , and Player II responds with  $V_n \in \tau_X^*$  such that  $x_n \in V_n \subseteq U_n$ .

Player II wins the play if  $\bigcap_{n=0}^{\infty} U_n \neq \emptyset$ . The space  $X$  is said to be *strongly  $\alpha$ -favorable* if Player II has a winning strategy in  $\text{Ch}(X)$ .

**Theorem 3.2 (G. Choquet, 1958).** *Let  $X$  be a metric space. Then  $X$  is completely metrizable if and only if  $X$  is strongly  $\alpha$ -favorable.*

*Proof.* Suppose first that  $X$  is complete with metric  $d$ . We define a strategy  $\tau$  for Player II. When Player I plays  $(x_n, U_n)$ , Player II chooses  $r_n > 0$  such that the open ball  $B(x_n, r_n) \subseteq U_n$ , with  $r_n < 2^{-n}$  and  $\overline{B(x_n, r_n)} \subseteq V_{n-1}$  (taking  $V_{-1} = X$ ). Player II then plays  $V_n = B(x_n, r_n)$ .

Since  $\text{diam}(V_n) \leq 2^{1-n} \rightarrow 0$  and  $\overline{V_n} \subseteq V_{n-1}$ , the Cantor intersection theorem shows that  $\bigcap_{n=0}^{\infty} \overline{V_n}$  consists of a single point  $x \in X$ . Since  $V_n \subseteq U_n$ , it follows that  $x \in \bigcap_{n=0}^{\infty} U_n$ . Thus  $\tau$  is a winning strategy for Player II, and  $X$  is strongly  $\alpha$ -favorable.

Conversely, suppose Player II has a winning strategy  $\tau$  in  $\text{Ch}(X)$ . By a standard result on complete regular spaces,  $X$  can be embedded as a dense subspace of its metric completion  $\tilde{X}$ . Using the strategy  $\tau$ , one constructs for each point of  $\tilde{X} \setminus X$  an open neighborhood in  $\tilde{X}$  that avoids the intersection of the played sets, showing that  $X$  is a  $G_\delta$ -subset of  $\tilde{X}$ . By Alexandrov's theorem, every  $G_\delta$ -subset of a complete metric space is completely metrizable. This completes the proof. 

### APPENDIX A. HISTORICAL REMARKS

**The Banach–Mazur and strong Choquet games.** The Banach–Mazur game is Problem 43 of the Scottish Book, posed by S. Mazur in 1935 [Mau81]: two players alternately choose nested intervals of the real line, and the first player wins if their intersection meets

a given set  $E$ . Mazur observed that the first player wins if  $E$  is comeager in some interval and that the second player wins if  $E$  is of the first category, and conjectured the converses. Banach recorded on 4 August 1935 that the conjecture is true, but his proof was never published; the first published proof, for arbitrary topological spaces, is Oxtoby's [Oxt57], to whom Theorem 2.2 is due. See [Tel87] for a survey.

The strong Choquet game goes back to Choquet's work on "espaces tamisables" [Cho58], and is expounded in his *Lectures on Analysis* [Cho69]; Theorem 3.2 is Choquet's theorem (see [Kec95, Section 8]).

**The notation  $[\mathbb{N}]^\omega$  and Hungarian combinatorics.** For a set  $S$  and a cardinal  $\kappa$ , one writes  $[S]^\kappa = \{X \subseteq S : |X| = \kappa\}$  and  $[S]^{<\kappa} = \{X \subseteq S : |X| < \kappa\}$ ; thus  $[\mathbb{N}]^\omega$  and  $[\mathbb{N}]^{<\omega}$  are the families of infinite and of finite subsets of  $\mathbb{N}$ , as opposed to the sequence spaces  $A^\omega$  and  $A^{<\omega}$  of Section 2. This is the notation of the partition calculus of P. Erdős and R. Rado [ER56], in which  $\kappa \rightarrow (\lambda)_\mu^r$  means: for  $|S| = \kappa$ , every partition of  $[S]^r$  into  $\mu$  pieces has a set  $H \subseteq S$  with  $|H| = \lambda$  and  $[H]^r$  inside a single piece. Ramsey's theorem becomes  $\omega \rightarrow (\omega)_k^r$  for finite  $r$  and  $k$ . For the infinite exponent the relation fails: using the axiom of choice, Erdős and Rado partitioned  $[\mathbb{N}]^\omega$  into two pieces with no infinite homogeneous set [ER52]. The theorems of Galvin–Prikry and Ellentuck [GP73, Ell74] recover  $\omega \rightarrow (\omega)_2^\omega$  for partitions that are Borel, or have the Baire property in the Ellentuck topology; in the companion notes on Ramsey theory, the Banach–Mazur game on the Ellentuck space is the bridge to these theorems. The notation, and the Budapest school of infinite combinatorics built around it, is set out in the monograph of Erdős, Hajnal, Máté, and Rado [EHMR84].

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